

Question 1

a)

I. Pulse 1 = Discrete quantitative variable

II. Ran = Nominal qualitative variable

III. Activity = Nominal qualitative variable

b) 123.8lb (pounds)

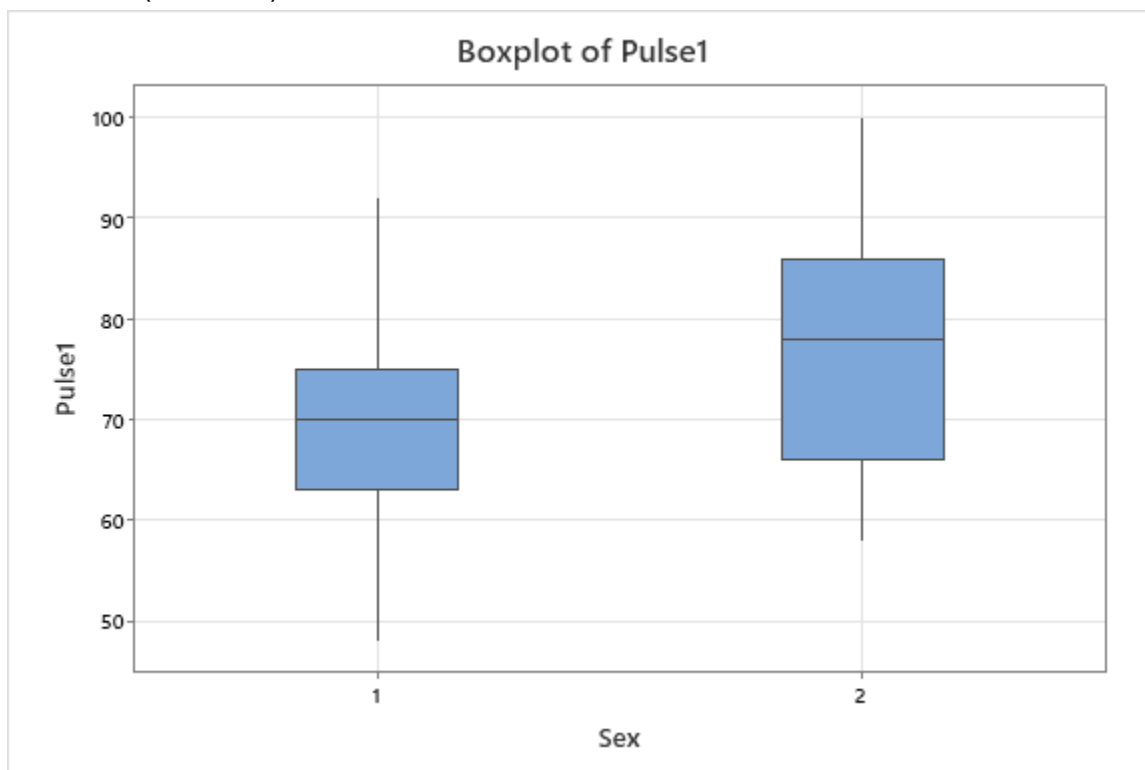
Reasoning, all female weight values divided by the number of values

c) 4

Reasoning, using the stats feature in Minitab and calculating all variables in this order, order the values, finding the median, finding the first and third Quartile Range, you can then calculate the IQR (Inter Quartile Range) of the height value for all Males (value 1 in sex table)

d) Sex = 1 (Males)

Sex = 2 (Females)

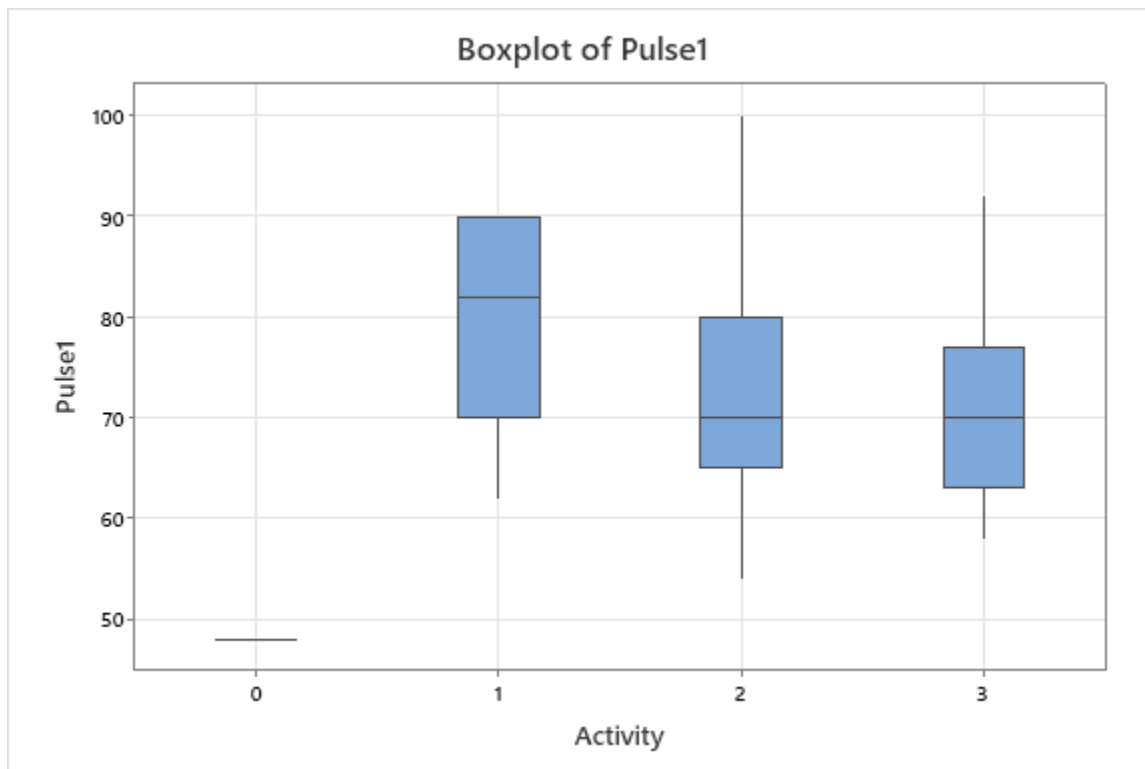


-Males have a considerably higher data consistency which also means the data is more

centralized (Less width span), This is represented by the IQR value (lower)

-Females have a less population (N= 35) which can create an argument of data reliability however if we look at the Median for females which is higher indicating right skewness (higher averages).

e) 1 = slight, 2 = moderate, 3 = a lot (Activity)



If we analyse Activity levels 2 and 3, we can rationalize those higher levels of activity and how they benefit our hearts pulsing rates (Heart stableness and Blood fluidity (lower pulsing)), this is provided for us with the IQR ranges, which are closely valued for 2 and 3 which is juxtaposed by Activity's 1 High IQR (larger data variability, less consistent heart pulsing).

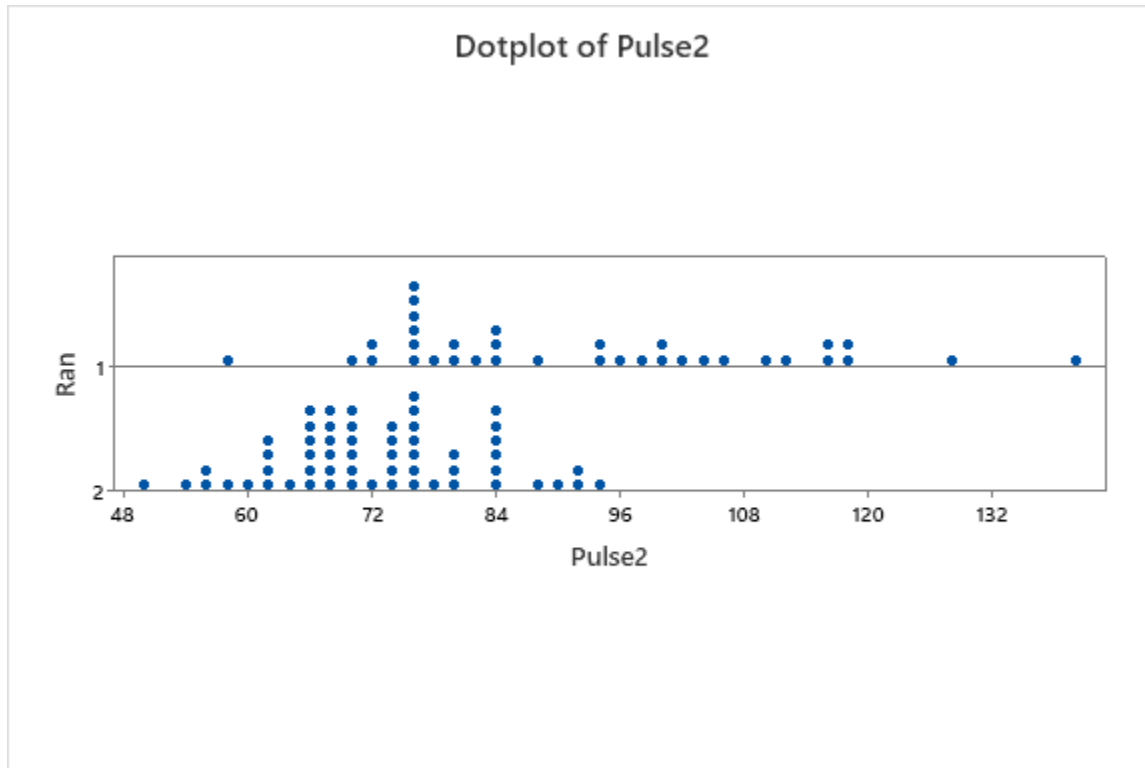
Theres also the fact that both activities 2 and 3 have the same median, we call these symmetrical data points, this suggest that when someone either moderately runs or runs a lot that in this case, they're central tendencies stays the same suggesting running until exertion does not continue to increase due to the bodies high fitness level (pulsing rates)

f) Regular smokers (value 1) = Mean Weight 152.286

Nonsmokers (value 2) = Mean Weight 142.031

By using Minitab I have concluded that Regular smokers tend to have higher weight values, this can be due to many lifestyles factors which may lead to smoking and higher weighted values.

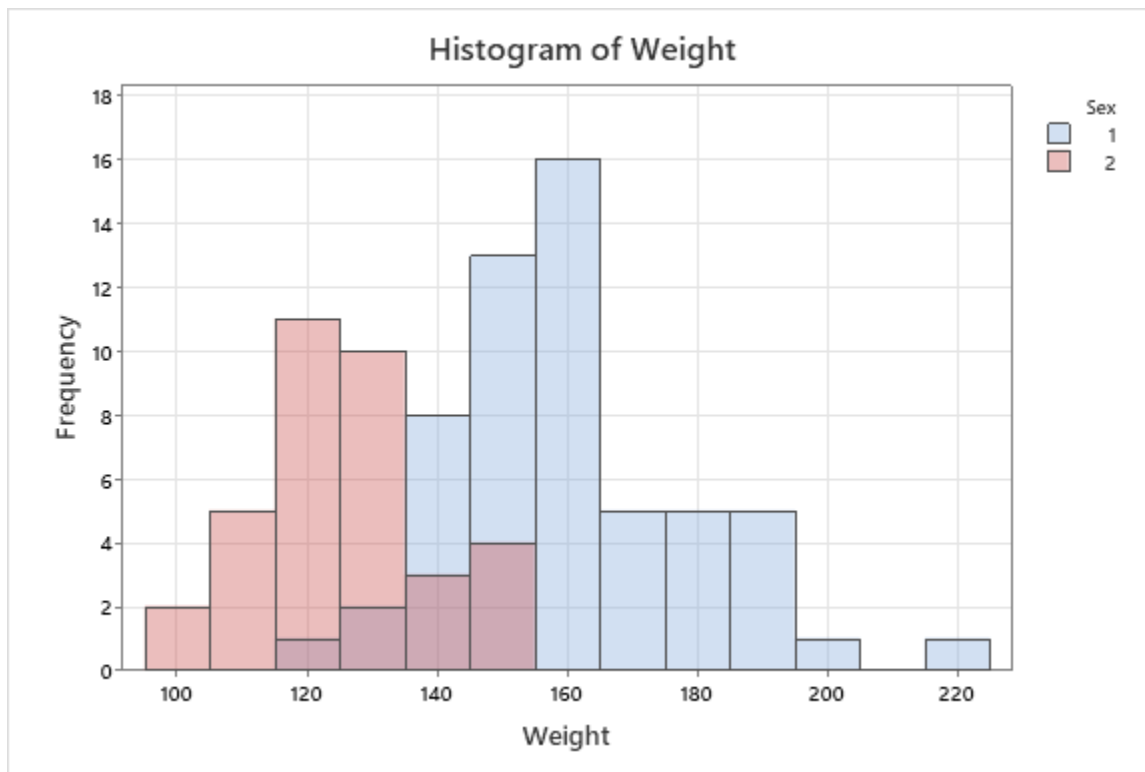
g) 1 = Ran in place, 2 = Did not run in place



-Values 1 had more data variability indicating a wider spread of data than Value 2, which suggests that the data is unpredictable. (higher tendencies)

-Values 2 which follows central tendencies following a narrower approach to displaying the data (median), suggesting data consistency!

h)



Question 2

a) This statement is incorrect since if something has the probability of 0.7 of happening it does not happen 'as often' as its opposite which inevitably would be 0.3 to make the probability of 1.

-70% has a higher probability of happening than 30%-

b) 52 cards

7 is our requirement

8 cards have been dealt and none have been 7

The probability of the next card being 7 (9th try):

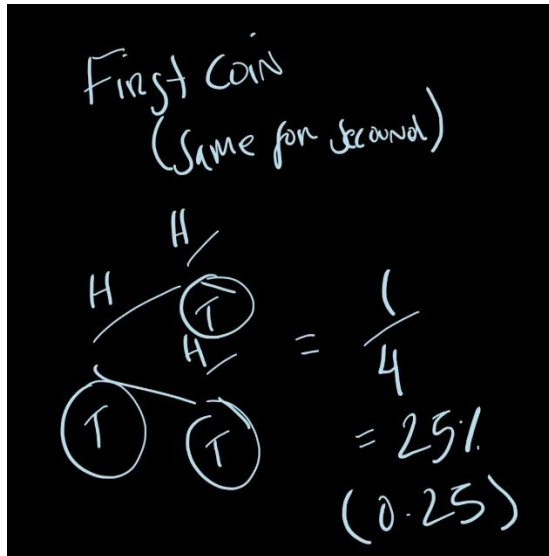
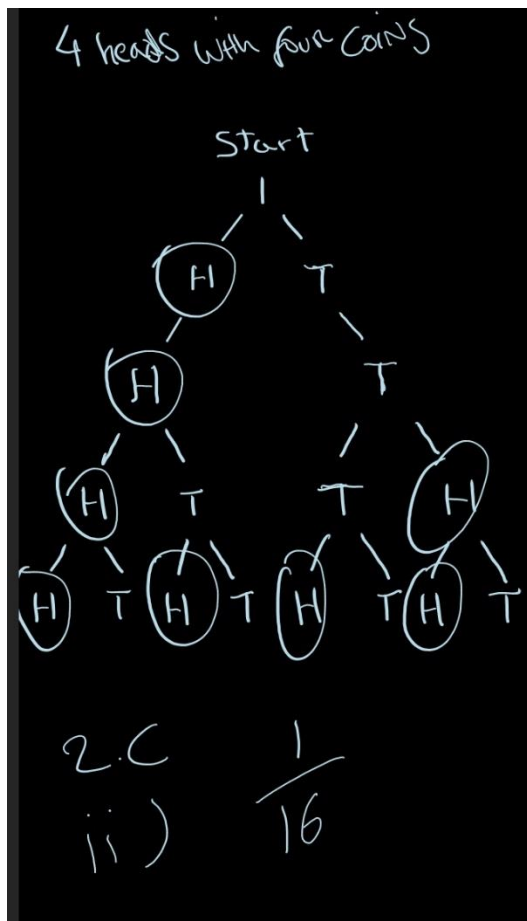
first: $52 - 8 = 44$

In this case there are 4, 7 remaining in the pack.

$4/44$ which in a simple form is $1/11$.

Answer: Therefore, the answer is that there is a $1/11$ chance of obtaining a 7 in the 9th try

c)

I. 0.25 ($\frac{1}{4}$)II. 0.0625 ($\frac{1}{16}$)

Question 3

a)

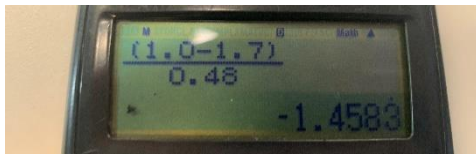


(formula)

$$Z = 1.708$$

$$z = 0.9625 \text{ (this is found through table 1)} \quad \text{Answer} = 0.9625 \times 100 = 96.25\%$$

b)



(both pictures are taken by me, with the given formula in the formula sheet book)



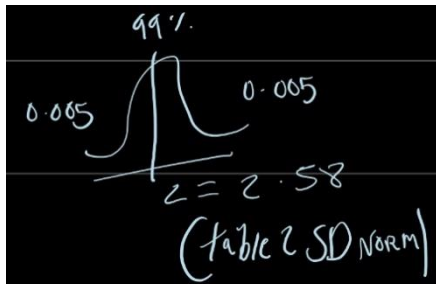
$$P(Z < 0.625) - P(Z < -1.4583)$$

$$= P(Z < 0.625) - (1 - P(Z < 1.4583))$$

$$= 0.7340 - 1 + 0.9276 = 0.6618 = 66.16\%$$

(This formula is also given in the sheet; the values are formed through the Stats Table which is within the subject area!)

c)



(This were done by done digitally)

$$(2.58 \times 0.48) + 1.7 = 2.9384 \text{ (Answer)}$$

$$Z = \frac{X - \mu}{\sigma} \text{ (formula sheet)}$$

d)

below	1.0 g/mile	no tax
between	1.0 and 1.6 g/mile	£50
between	1.6 and 2.2 g/mile	£100
between	2.2 and 2.8 g/mile	£250
above	2.8 g/mile	£2000

15 million petrol vehicles in the UK (sample)

Emission Level	Proportion	Number of cars	Tax level (£)	Total tax payable (£)
1.0	0.924845	x15M=cars	£0	(0x0) =£0
1.0 to 1.6	0.5826	x15M=cars	£50	Cars x 50=£436,950,000
1.6 to 2.2	0.8510	x15M=cars	£100	Cars x 100=1,276,500,000
2.2 to 2.8	0.9891	x15M=cars	£250	Cars x 250=3,709,125,000
Above 2.8	N/A	x15M=cars	£2000	Cars x 2000=

$$\begin{array}{r}
 1.0 - 1.7 \\
 \hline
 0.924845 \quad 0.48 \\
 = -1.4583
 \end{array}$$

$$\begin{array}{r}
 1.6 - 1.7 \\
 \hline
 0.5826 \quad 0.48 \\
 = -0.2083
 \end{array}$$

$$\begin{array}{r}
 2.2 - 1.7 \\
 \hline
 0.8510 \quad 0.48 \\
 = 1.0416
 \end{array}$$

$$\begin{array}{r}
 2.8 - 1.7 \\
 \hline
 0.9891 \quad 0.48 \\
 = 2.2916
 \end{array}$$

Question 4

Claim- 60% of people wanted a regulation ban to mobile phone usage on public transport.

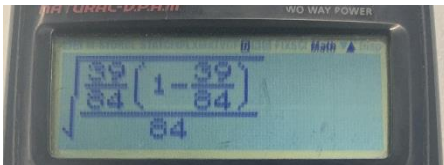
84 people, randomly selected

39 agreed on the regulation ban

- a) 99% confidence interval of people agreeing to the regulation ban

$$\hat{p} \pm z \times \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$$

(this photo is from the Stat4002 Formula sheet)



(this photo was taken by me)

$$0.054 = 5.4\%$$

$$\text{For a 99\%} = (39/84 - 2.58 \times 0.054, 39/84 + 2.58 \times 0.054)$$

$$\text{Answer} = (0.3249657143, 0.6036057143)$$

$$\text{which can be converted into \%} = (32.50\%, 60.40\%)$$

- b) Based on evidence found, we reject any alternative hypothesis and stay with the null hypothesis for this example.